TECHNICAL DATA SHEET

Weld-On 3 DCM TCE FREE

Acrylic Plastic Cement

FOR INDUSTRIAL AND PROFESSIONAL USE ONLY. NOT FOR CONSUMER USE.

SUBSTRATE RECOMMENDATIONS

Weld-On® 3 DCM TCE FREE is a methylene chloride and trichloroethylene free solvent cement, formulated for cementing acrylic (polymethylmethacrylate) plastic to itself. It is not recommended for cross-linked acrylic.

BONDING RECOMMENDATIONS

Weld-On 3 DCM TCE FREE is used extensively in sign fabrication. Strong butt joints are made with flat sheets by using the soak method for such applications as cementing letters to acrylic panels. Butyrate trim-capping of acrylics is another highly recommended use for Weld-On 3 DCM TCE FREE. It is also widely used in the POP industry (e.g. display), and in the fabrication and bonding of plastic containers, housewares, bottles and many other applications using the capillary method.

GENERAL DESCRIPTION

Weld-On 3 DCM TCE FREE is a water-thin, flammable cement formulated to quickly develop high strength, clear bonds to many thermoplastic substrates, particularly acrylic. The bond is achieved by first softening the surfaces to be joined and then fusing them together with dissipation of the solvent. The initial bond forms within a few minutes and quickly increases in strength. Substantial strength will be obtained within only a few hours. Under certain conditions, especially with high humidity, a white mark on the plastic (commonly called blushing) may sometimes be encountered. Where a more blush resistant cement is required, Weld-On 4 DCM TCE FREE is recommended.

TYPICAL BOND STRENGTH†

Substrate Material	Aged Bond Strength, lbs/in² (kg/cm²)		
	2 Hours	24 Hours	1 Week
Acrylic (Cast & Extruded)	732 (51.5)	2,825 (198.6)	2,924 (205.6)

†Substrate thickness: 0.25 inch (0.64 cm). Bond area: 1.0² in (6.45²cm)

ADHESIVE PROPERTIES AND CHARACTERISTICS @ 73°F (23°C)

Color:	Clear	Fixture Time:	2 Minutes
Viscosity:	Water Thin	Time to Reach 65-80% of Ultimate Bond Strength:	24 Hours
Working Time:	1 Minutes	Specific Gravity:	1.218 ± 0.01

DIRECTIONS FOR USE

GENERAL: Surfaces to be joined must be clean, dry, and fit intimately without forcing. Apply cement with syringe, eyedropper or brush. Assemble while parts are still wet. If cement is applied to one surface, let the two surfaces be in gentle contact for a few seconds to allow the cement to soften the dry surfaces, then press parts together in firm contact. Initial bonds form very quickly. 90-96% of the ultimate bond strength will be obtained within 24 hours. Strength will continue to increase for several weeks.

CAPILLARY METHOD: Parts are placed lightly together and cement is applied to the edge of the joint with syringe or eyedropper. By capillary action, the cement will flow a considerable distance, approximately 0.25 inch (0.64 cm), between two such surfaces. Allow a few seconds for the cement to soften the surfaces. Press parts firmly together.

SOAK METHOD: Vertically dip surfaces until softened (approximately 2 to 5 minutes), then join pieces firmly together.

- In many cases, this cement dries too fast to apply by brush to the parts before they are joined.
- If crazing is a problem, we suggest you consider annealing before cementing.

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AVAILABILITY

This product is available in gallon (3.785 liters), quart (946 ml), pint (473 ml) and 1/4 pint (118 ml) metal cans. For detailed information on containers and applicators, refer to the current Sign and Display Assembly Selection Guide and Price List.

SHELF LIFE

Two years in tightly sealed containers. The date code of manufacture is stamped on the bottom of the container. Stability of the product is limited by the evaporation of the solvent when the container is opened. Evaporation of solvent will cause the cement to thicken and reduce its effectiveness. Adding of thinners to change viscosity is not recommended and may significantly change the properties of the cement.

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QUALITY ASSURANCE

Weld-On® 3 DCM TCE FREE is carefully evaluated to assure that consistent high quality is maintained. Fourier transform infrared spectroscopy, gas chromatography, and additional in depth testing ensures each batch is manufactured to exacting standards. A batch identification code is stamped on each can and assures traceability of all materials and processes encountered in manufacturing this plastic cement for its intended specific application.

SHIPPING

Shipping Information for One Liter and Above: Proper Shipping Name: Adhesives. Hazard Class: 3. Identification Number: UN 1133.
Packing Group: II. Label Required: Class 3 Flammable Liquid.
Shipping Information for Less than One Liter: Proper Shipping Name: Consumer Commodity. Hazard Class: ORM-D.

SAFETY AND ENVIRONMENTAL PRECAUTIONS

This product is a flammable, very fast evaporating solvent cement. It is considered a hazardous material. In conformance with the Federal Hazardous Substance Labeling Act, the following hazards and precautions are given. Purchasers who may re-package this product must also conform to all local, state, and federal labeling, safety and other regulations.

DANGER: HIGHLY FLAMMABLE LIQUID AND VAPOR. HARMFUL IF INHALED. MAY CAUSE AN ALLERGIC SKIN REACTION. SUSPECTED OF CAUSING CANCER.

Keep out of the reach of children. Do not take internally. Keep away from heat, spark, open flame and other sources of ignition. Contact with hot surfaces may produce toxic effects. Keep container closed when not in use. Store away from direct sunlight below 80º F (27º C). Use only in adequate ventilation. Avoid breathing of vapors. Atmospheric levels should be maintained below established exposure limit values. See Sections II and VIII of Safety Data Sheet. If airborne concentrations exceed these limits, use a supplied air respirator. Do not use a chemical cartridge respirator. For emergencies and other conditions where short-term exposure may be exceeded, use an approved positive pressure self-contained breathing apparatus. In confined areas, use a positive pressure self-contained breathing apparatus (SCBA). Do not smoke, eat or drink while working with this product. Avoid contact with skin, eyes, and clothing. May cause eye injury. Protective equipment such as gloves, safety goggles, and impervious apron should be used. Carefully read Safety Data Sheet and follow all precautions.

Contains trans-dichloroethylene (156-60-5), methyl methacrylate (80-62-6) and Nitromethane (75-52-5) . Do not use this product for other than intended use.

“Title III Section 313 Supplier Notification:” This product contains toxic chemicals subject to the reporting requirements of Section 313 of the Emergency Planning and Community Right-to-Know Act of 1986 and of 40CFR372. This information must be included in all I Safety Data Sheets that are copied and distributed for this material.

FIRST AID

Inhalation: If overcome with vapors, remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Call physician.
Eye Contact: Flush with plenty of water for 15 minutes and call a physician.
Skin Contact: Wash skin with plenty of soap and water for at least 15 minutes. If irritation develops, get medical attention.
Ingestion: If swallowed, give 1 or 2 glasses of water or milk. Do not induce vomiting. Contact physician or poison control center immediately.

IMPORTANT NOTE

THIS PRODUCT IS INTENDED FOR INDUSTRIAL AND PROFESSIONAL USE ONLY. NOT FOR CONSUMER USE. These suggestions and data are based on information we believe to be reliable. Users should verify by test that this product, as well as these methods, is suited to their applica-

WARRANTY

IPS Corporation (IPS Corp.) warrants that all new IPS Corp. products shall be of good quality and free from defects in material and workmanship for the shelf life as indicated on the product. If any IPS Corp. product becomes defective, or fails to conform to our written limited warranty under normal use and storage conditions, then IPS Corp. will, without charge, replace the nonconforming product. However, this limited warranty shall not extend to, nor shall IPS Corp. be responsible for, damages or loss resulting from accident, misuse, negligent use, improper application, or incorporation of IPS Corp. products into other products. In addition, any repackaging of IPS Corp. products also shall void the limited warranty. IPS Corp. shall not be responsible for, nor does this limited warranty extend to, consequential damage, or incidental damage or expense, including without limitation, injury to persons or property or loss of use. Please refer to our standard IPS Corp. Limited Warranty for additional provisions.

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